

Quadruped Robot Mechanical Design

A comprehensive mechanical engineering report presenting the preliminary design of a lightweight, stable, and educationally focused quadruped robot – covering chassis specifications, leg architecture, joint kinematics, motor and battery selection, torque analysis, stability principles, and proposed walking gait.

Prepared By: Ebtihal Fawaz Alamri

Major: Computer and Network Engineering

University: University of Jeddah

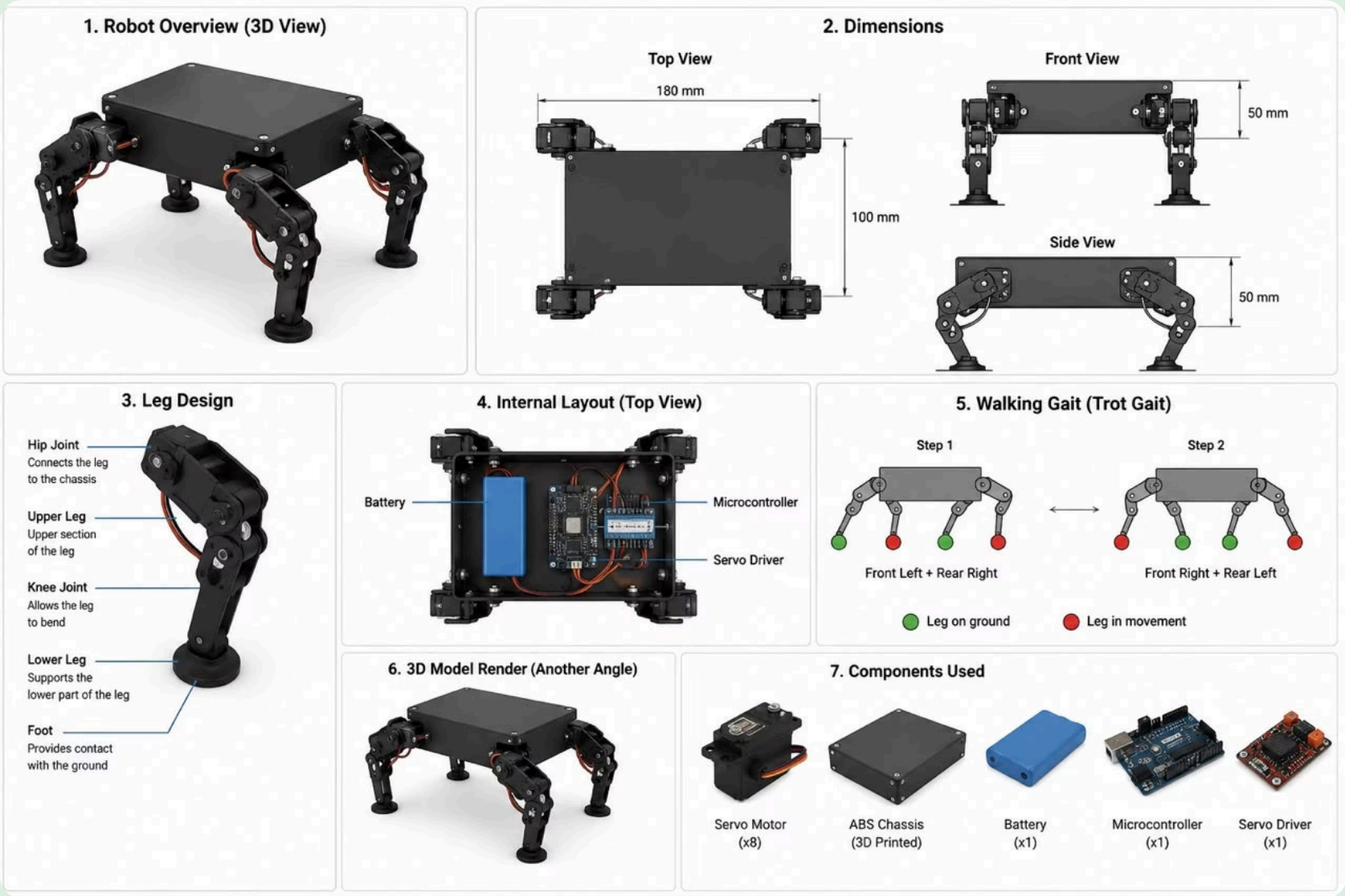
Program: Smart Methods Robotics Training

Table of Contents

This report presents the complete preliminary mechanical design of a quadruped robot intended for educational applications. The document is structured to guide the reader through each engineering decision – from chassis and leg design through to motor selection, torque calculations, stability analysis, and gait planning. Each section builds upon the previous to form a cohesive and well-justified design proposal.

01	Project Objective Overview of the robot's purpose, design goals, and intended educational applications. Defines the scope of the mechanical design and the principles applied throughout the project.	02	Body and Chassis Specifications Material selection, dimensions, manufacturing method, and structural rationale for the robot chassis. Includes justification for the use of ABS plastic and 3D printing technology.
03	Leg Design Component breakdown of each leg, including hip joint, upper leg, knee joint, lower leg, foot, and rubber foot pad assembly.	04	Joints and Degrees of Freedom (DOF) Analysis of joint configuration, total degrees of freedom, and the mechanical flexibility this provides for stable walking.
05	Motor and Battery Selection Justification for servo motor selection, battery type, and component specifications. Includes internal servo motor composition and battery placement rationale.	06	Preliminary Torque Calculation Step-by-step force and torque analysis for the hip joint, including safety factor application and required minimum torque determination.
07	Stability and Center of Gravity (COG) Design strategies for maintaining robot stability during operation, including weight distribution and component placement.	08	Proposed Walking Gait Description of the Trot gait, diagonal leg coordination, and advantages for balance and power efficiency.
09	Expected Mechanical Problems Anticipated mechanical issues during operation and their proposed engineering solutions.	10	Design Assumptions and Conclusion Summary of all design assumptions made throughout the project and the overall conclusions drawn from the mechanical design process.

 **Figure 1** – Overall Mechanical Design of the Proposed Quadruped Robot. The figure presents the complete mechanical design, including the 3D model, chassis dimensions, leg structure, internal component layout, walking gait, and the main components used.



Project Objective

The primary objective of this project is to design a simple quadruped robot capable of standing and walking while applying the fundamental principles of mechanical design, stability, joint movement, and motor selection. The robot is intended for educational purposes and is designed to be lightweight, stable, and easy to manufacture using 3D printing technology. This project serves as a practical application of core mechanical engineering concepts, providing hands-on experience in the design and development of a functional robotic system.

The design process encompasses all major mechanical aspects of a quadruped robot, including the selection of appropriate materials for the chassis, the structural design of each leg, the configuration of joints and degrees of freedom, the selection of servo motors and battery, torque calculations with safety factors, stability analysis through center of gravity management, and the selection of an appropriate walking gait. Each of these elements is addressed in detail in the following sections of this report, forming a complete preliminary mechanical design suitable for educational implementation and further development.

Design Goals

- Lightweight construction for efficient operation
- Stable standing and walking capability
- Simple and cost-effective manufacturing via 3D printing
- Educational applicability for robotics training

Engineering Principles Applied

- Mechanical design and structural analysis
- Stability and center of gravity management
- Joint movement and degrees of freedom
- Motor selection and torque calculation

The quadruped robot design applies fundamental principles of mechanical engineering to create a functional and educationally valuable platform. By focusing on simplicity and manufacturability, the design ensures that the robot can be produced using accessible 3D printing technology while still demonstrating key concepts in robotics such as joint kinematics, motor control, stability, and gait coordination. The result is a well-documented preliminary design that provides a solid foundation for future implementation and further development.

Body and Chassis Specifications

The robot chassis is designed to provide a lightweight, stable, and durable structure suitable for educational applications. A rectangular chassis was selected because it provides a simple and stable structure while offering sufficient space for the battery, controller, and servo motors. The chassis dimensions and overall robot design are illustrated in Figure 1. The main specifications of the chassis are detailed in the table below.

Property	Specification
Body Shape	Rectangular
Material	ABS Plastic
Manufacturing Method	3D Printing
Thickness	3 mm
Dimensions	180 mm × 100 mm × 50 mm
Estimated Weight	Approximately 1 kg
Number of Legs	Four (4)

Material Selection — ABS Plastic

ABS plastic was chosen because it offers better heat resistance and durability than PLA, making it suitable for robotic applications. The material properties of ABS ensure that the chassis can withstand the mechanical stresses encountered during robot operation, including joint actuation forces and ground contact impacts. The improved heat resistance of ABS compared to PLA is particularly important given the thermal output of servo motors and the battery during extended operation.

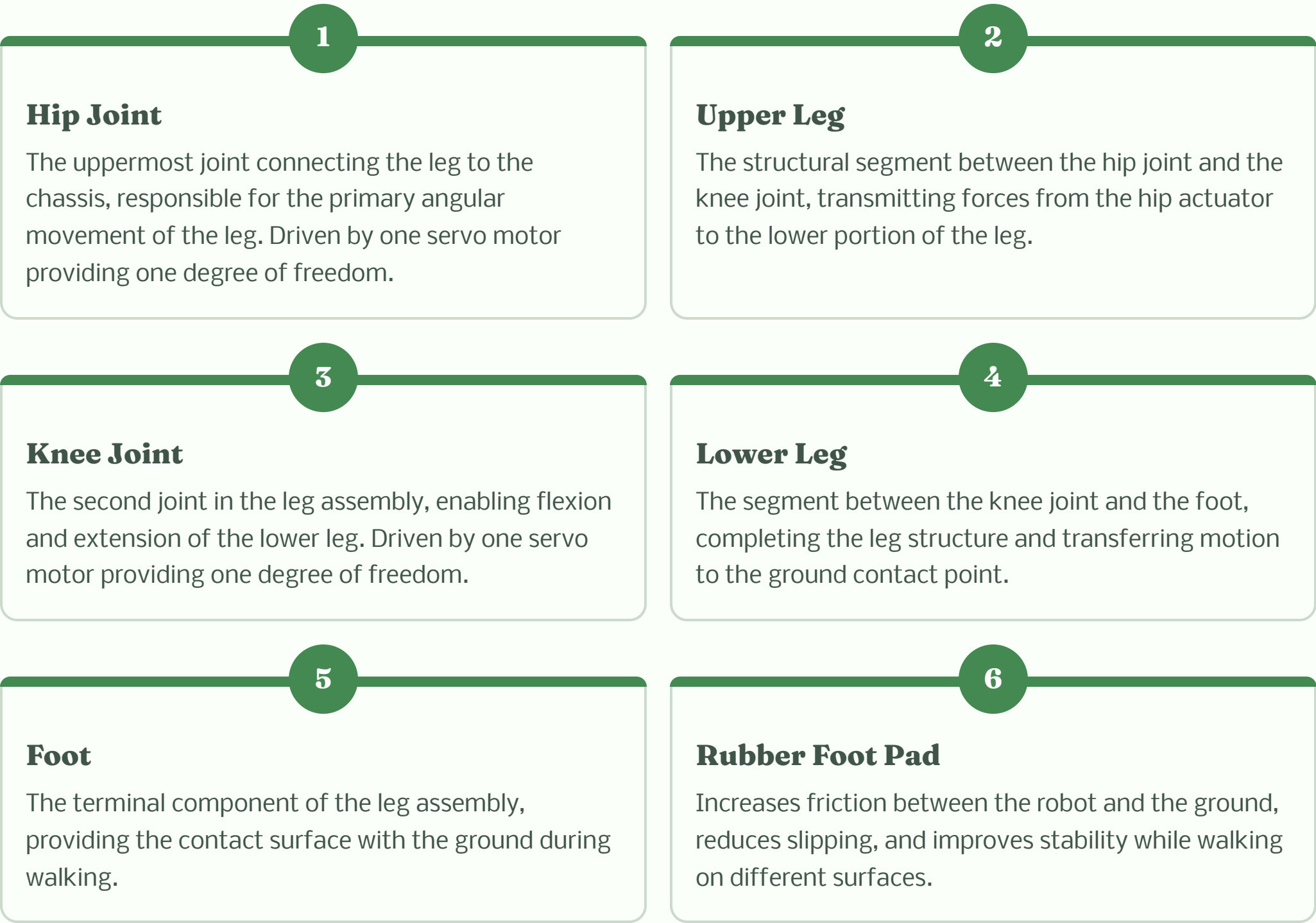
The selected dimensions of 180 mm × 100 mm × 50 mm provide enough space for all components – including the battery, controller, and servo motors – while maintaining good balance and stability during operation. The estimated total weight of approximately 1 kg ensures that the robot remains lightweight enough for the selected servo motors to drive effectively while still being substantial enough to maintain stability during walking. The rectangular body shape further contributes to a predictable and manageable center of gravity, which is essential for stable quadruped locomotion.

Manufacturing Method — 3D Printing

The chassis is manufactured using 3D printing, which enables accurate fabrication, lightweight construction, easy customization, and cost-effective prototyping. The 3 mm wall thickness was selected to provide adequate structural rigidity while minimizing overall weight. This manufacturing approach also allows for rapid iteration and modification of the chassis design as the project progresses, which is especially valuable in an educational context where design improvements are expected.

Leg Design and Degrees of Freedom

Each robot leg consists of multiple structural and functional components that work together to enable stable walking motion. The leg structure is illustrated in Figure 1, showing the hip joint, knee joint, lower leg, and foot assembly. The design of each leg balances mechanical simplicity with the functional requirements of quadruped locomotion, ensuring that the robot can achieve stable walking while remaining straightforward to manufacture and assemble.



JJIIJoints and Degrees of Freedom (DOF)

Each leg contains two joints: one hip joint and one knee joint. With four legs on the robot, the total joint count and degrees of freedom are calculated as follows:

2	8	8
Joints per Leg	Total Joints	Total DOF
1 Hip Joint + 1 Knee Joint per leg	2 × 4 legs = 8 total joints	Each joint provides 1 DOF; 2 × 4 = 8 DOF

This configuration provides sufficient flexibility for stable walking while maintaining a simple mechanical design. Each joint is controlled independently by one servo motor, allowing precise angular positioning at both the hip and knee of each leg. The 8-DOF configuration is well-suited for educational applications, as it demonstrates the fundamental concepts of joint kinematics and degrees of freedom in a quadruped robot without introducing excessive mechanical complexity. The joints and degrees of freedom of the robot are shown in Figure 1.

Motor and Battery Selection

Servo motors were selected to drive each joint of the quadruped robot because they provide accurate angular positioning, fast response, and repeatable movement. Each joint is driven by one servo motor, resulting in a total of eight servo motors for the complete robot. The selection of servo motors is based on their ability to meet the torque requirements calculated for the hip joint while providing the precision and reliability needed for stable quadruped locomotion.

Servo Motor Internal Components

Each servo motor consists of the following internal components that work together to provide precise angular control:

- DC Motor – provides rotational power
- Gear Train – increases torque output
- Position Sensor – monitors angular position
- Control Circuit – processes control signals

Total Servo Motors = **8** (one per joint)

Why Servo Motors?

- Precise angle control for accurate joint positioning
- High positioning accuracy for repeatable movement
- Fast response to control signals
- Reliable operation under repeated use
- Suitable for quadruped robot joint applications

Based on the torque calculation, each servo motor should provide at least **0.36 N·m** of torque. This value was determined through preliminary torque analysis of the hip joint, including the application of a safety factor of 3 to account for dynamic loads during walking. Servo motors meeting or exceeding this torque specification, such as the MG996R or equivalent, are recommended for this design.

Battery Selection

Battery Type: Lithium Polymer (Li-Po)

A Lithium Polymer battery was selected because it is lightweight, has high energy density, and can provide sufficient current for multiple servo motors operating simultaneously. The high energy density of Li-Po batteries ensures that the robot can operate for a reasonable duration without adding excessive weight to the chassis. The lightweight nature of Li-Po batteries is particularly important for a quadruped robot, as additional battery weight directly increases the load on each servo motor and affects overall stability.

The battery is positioned near the center of the chassis to improve weight distribution and enhance robot stability. Central placement of the battery helps maintain the center of gravity near the middle of the chassis, reducing the possibility of tipping during movement. This placement also ensures that the weight of the battery is evenly distributed across all four legs, contributing to balanced operation during walking.

Preliminary Torque Calculation

A preliminary torque calculation was performed for the hip joint to determine the minimum torque requirement for each servo motor. The calculation considers the estimated load on one leg, the distance from the hip joint to the center of mass, and a safety factor to account for dynamic loads during walking. The step-by-step analysis is presented below.

Selected Joint and Assumptions

Joint: Hip Joint

- Estimated load on one leg = **0.2 kg**
- Distance from hip joint to center of mass = **0.06 m**

Force Calculation

Using Newton's second law ($F = m \times g$):

$$F = 0.2 \times 9.81$$

$$F = 1.96 \text{ N}$$

Torque Calculation

Using the torque formula ($T = F \times d$):

$$T = 1.96 \times 0.06$$

$$T \approx 0.12 \text{ N}\cdot\text{m} \text{ (approximate static torque)}$$

Safety Factor Application

During walking, additional loads occur due to ground impact, friction, and dynamic motion. Therefore, a **Safety Factor of 3** is applied:

$$\text{Required Torque} = 0.12 \times 3$$

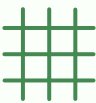
$$\text{Required Torque} = 0.36 \text{ N}\cdot\text{m}$$

- ❑ The recommended minimum torque for each servo motor is **0.36 N·m**, ensuring reliable operation during walking. This value accounts for static loading as well as dynamic effects including ground impact forces, friction at the foot-ground interface, and inertial loads generated during leg movement. Servo motors selected for this design must meet or exceed this torque specification to ensure safe and reliable operation throughout the robot's walking cycle.

Stability, Center of Gravity and Walking Gait

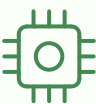
Stability and Center of Gravity (COG)

Robot stability is achieved through a combination of design strategies that work together to maintain balance during operation. The center of gravity is located near the middle of the chassis, reducing the possibility of tipping during movement. The stability concept and center of gravity (COG) are illustrated in Figure 1. The following design features contribute to overall robot stability:



Leg Distribution

Evenly distributing the four legs around the chassis provides a stable base of support, ensuring that the robot's weight is balanced across all contact points with the ground during both standing and walking.



Component Positioning

Positioning the electronic components centrally within the chassis further contributes to balanced weight distribution, ensuring that no single area of the robot is disproportionately heavy.

The use of a lightweight ABS chassis further supports stability by minimizing the overall mass that must be controlled during walking. The combination of these design strategies ensures that the robot maintains a low and centered center of gravity, which is essential for stable quadruped locomotion.



Battery Placement

Placing the battery near the center of the chassis improves weight distribution and keeps the center of gravity close to the geometric center of the robot, reducing the tendency to tip during leg movement.



Rubber Foot Pads

Adding rubber foot pads improves traction between the robot's feet and the ground, reducing slipping and enhancing stability while walking on different surfaces.

Proposed Walking Gait

The robot uses the **Trot Gait**, in which diagonal legs move together in coordinated pairs. This gait is well-suited for the 8-DOF configuration of this robot and provides a good balance between stability, smoothness, and implementation simplicity.

Diagonal Leg Pairs

- Front Left + Rear Right (move together)
- Front Right + Rear Left (move together)

The Trot gait sequence is illustrated in Figure 1, showing the coordinated movement of the diagonal legs.

Advantages of the Trot Gait

- Good balance during walking
- Smooth walking motion
- Easy implementation with 8-DOF configuration
- Efficient power consumption

Expected Mechanical Problems and Design Assumptions

Expected Mechanical Problems

During the operation of the quadruped robot, several mechanical problems may occur. The expected issues and their proposed solutions are listed below. Identifying these potential problems in advance allows for proactive design decisions that mitigate their impact on robot performance and longevity.

Problem	Proposed Solution
Servo Motor Overload	Select servo motors with sufficient torque to handle the required load safely. The preliminary torque calculation determined a minimum requirement of 0.36 N·m per motor, and motors meeting or exceeding this specification should be used.
Joint Wear	Use durable joint materials and perform regular maintenance to reduce wear over time. Proper material selection for joint components will extend the operational life of the robot.
Balance Instability	Keep the center of gravity near the center of the chassis and distribute the robot's weight evenly. Battery placement near the chassis center and central positioning of electronic components are key design measures.
Chassis Deformation Due to Heat	Use ABS plastic, which offers better heat resistance than PLA, and provide proper battery protection to minimize heat-related deformation of the chassis structure.
High Power Consumption	Use an efficient walking gait – the Trot gait was selected for its efficient power consumption – and select a suitable Li-Po battery with sufficient energy density to support extended operation.

Design Assumptions

The following design assumptions were made throughout the development of this preliminary mechanical design. These assumptions define the scope and constraints of the project and should be considered when interpreting the design specifications and calculations presented in this report.

- The robot is designed for **educational purposes**.
- The chassis is made of **ABS plastic** and manufactured using **3D printing** technology.
- Each leg has **two joints** (hip and knee), and each joint is controlled by **one servo motor**, for a total of 8 servo motors.
- The robot uses a **Lithium Polymer (Li-Po) battery** positioned near the center of the chassis for improved weight distribution.
- The robot uses the **Trot Gait** for walking, with diagonal leg pairs moving in coordination.
- **Rubber foot pads** are used to improve traction and walking stability on different surfaces.

Conclusion and References

Conclusion

This report presented the preliminary mechanical design of a quadruped robot for educational applications. The proposed design combines a lightweight ABS chassis, eight servo motors, a Li-Po battery, optimized torque calculations, and a Trot gait to achieve stable and efficient walking performance. Each aspect of the design – from chassis material selection and leg architecture to joint configuration, motor selection, torque analysis, stability management, and gait planning – has been carefully considered and justified based on fundamental mechanical engineering principles.

The chassis design utilizes ABS plastic manufactured via 3D printing, providing a lightweight yet durable structure with dimensions of 180 mm × 100 mm × 50 mm and an estimated weight of approximately 1 kg. Each of the four legs incorporates two joints (hip and knee), resulting in a total of 8 degrees of freedom driven by 8 servo motors. Preliminary torque calculations for the hip joint determined a minimum required torque of 0.36 N·m per motor after applying a safety factor of 3 to account for dynamic loads during walking. Stability is maintained through even leg distribution, central battery placement, and rubber foot pads for improved traction. The Trot gait was selected for its balance, smoothness, ease of implementation, and efficient power consumption.

The design addresses potential mechanical problems including servo motor overload, joint wear, balance instability, chassis deformation due to heat, and high power consumption through proactive engineering solutions. The design provides a solid foundation for future implementation and further development, offering a practical platform for applying and demonstrating core concepts in mechanical design, stability, joint movement, and motor selection within an educational robotics context.

References

Reference 1

Boston Dynamics. *Quadruped Robot Design Concepts*.

Reference 2

Servo Motor Datasheets (MG996R or equivalent).

Reference 3

Smart Methods Robotics Training Materials.